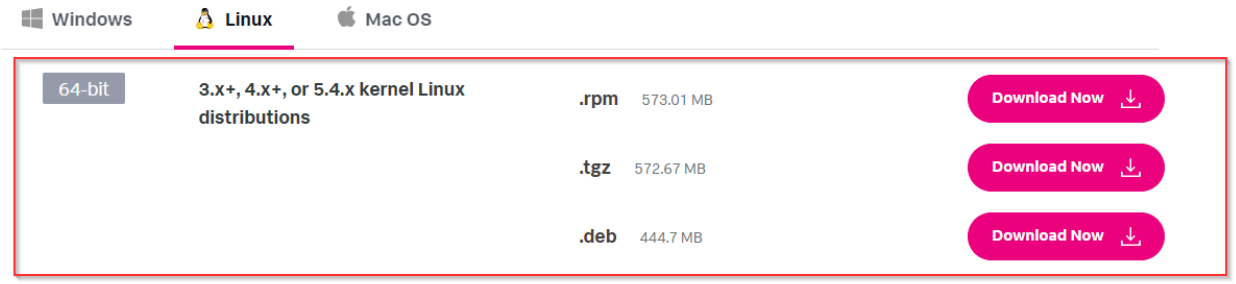
SPLUNK - Setup Lab & Filters (Linux)

Installing On Linux System

Create an account on splunk.com and go to this Splunk Enterprise download link to select the installation package for the latest version.



STEP 1: Become a sudo user

```
ubuntu@coffely:~/Downloads/splunk/$ ls splunk_installer.tgz
splunkforwarder.tgz

ubuntu@coffely:~/Downloads/splunk/$ sudo su
root@coffely:~/Downloads/splunk/$
```

STEP 2: Splunk Installation

Splunk installation is as simple as running a command. You will need to uncompressed Splunk by running the following command.

```
root@coffely:~/Downloads/splunk/$ tar xvzf splunk_installer.tgz
splunk/
splunk/splunk-9.0.3-dd0128b1f8cd-linux-2.6-x86_64-manifest
splunk/swidtag/
splunk/swidtag/splunk-Splunk-Enterprise-primary.swidtag
splunk/ftr
splunk/openssl/
....
....
....
splunk/etc/splunk-enttrial.lic
splunk/etc/splunk-launch.conf.default
splunk/etc/findlogs.ini
splunk/etc/log-cmdline.cfg
splunk/etc/deployment-apps/
splunk/etc/deployment-apps/README
```

```
splunk/etc/searchLanguage.xml
splunk/etc/log-debug.cfg
splunksetup
```

After the installation is complete, a new folder named `splunk` will be created, as shown below. Let's now move this folder to the `/opt/` directory and start working on Splunk from there.

Splunk setup

```
root@coffely:~/Downloads/splunk/$ ls
splunk splunk_installer.tgz splunkforwarder.tgz
root@coffely:~/Downloads/splunk/$ mv splunk /opt/
```

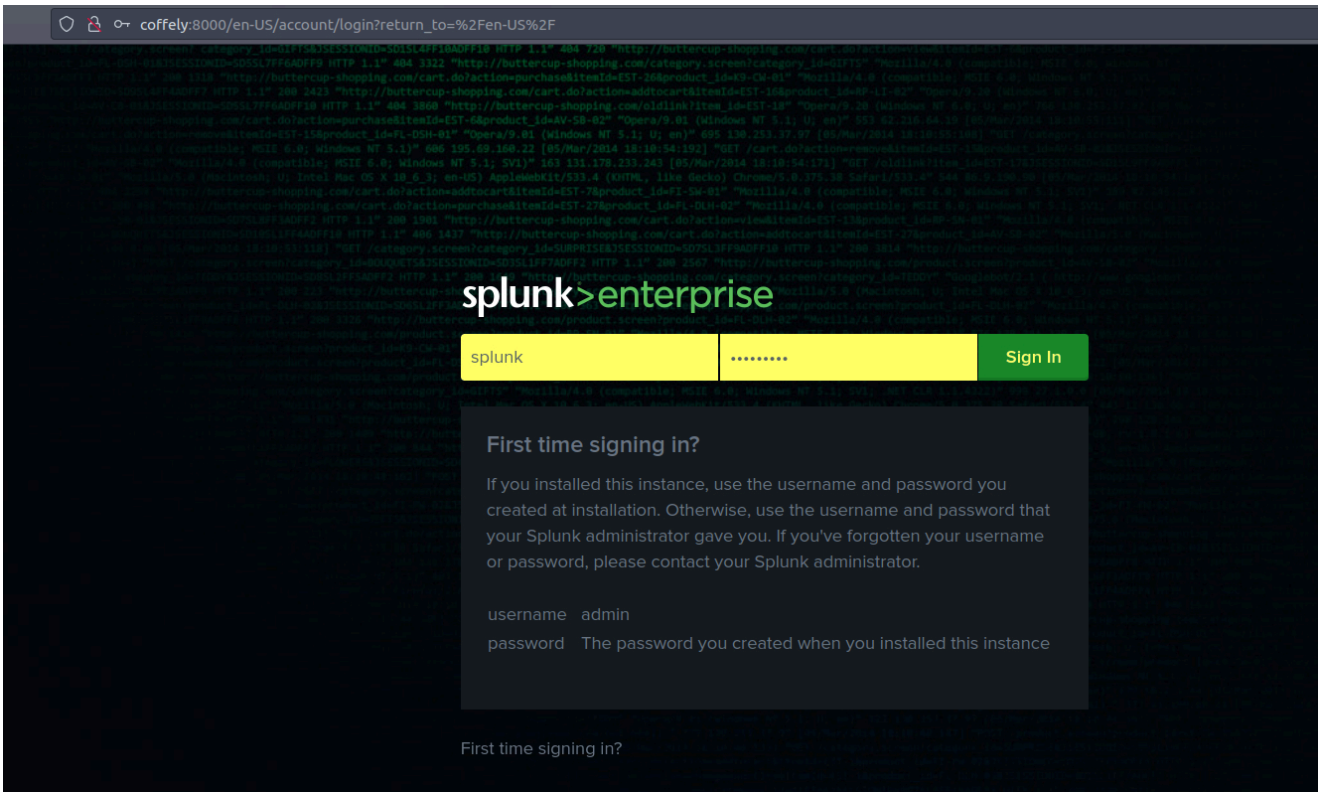
Starting Splunk

The above step unzips the Splunk installer and installs all the necessary binaries and files on the system. Once installed, go to the directory `/opt/splunk/bin` and run the following command to start Splunk `./splunk start --accept-license`. As it is the first time we are starting the Splunk instance, it will ask the user for admin credentials. Create a user account and proceed.

```
root@coffely:~/Downloads/splunk/# cd /opt/splunk/bin
root@coffely:/opt/splunk/bin#./splunk start --accept-license
This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup.
Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: splunkadmin
Password must contain at least:
   * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
....
....
....
Waiting for web server at http://127.0.0.1:8000 to be
available............... Done


If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://coffely:8000
```

Accessing Splunk

Congrats! - We successfully installed Splunk on our Linux machine, which took us less than 10 minutes. To access Splunk, open the browser within the VM and go to the address `http://coffely:8000`. If you are connected to the VPN, you can access Splunk right in your browser by going to the address. `http://10.49.186.69:8000`.



Command: splunk start

The `splunk start` command is used to start the Splunk server. This command starts all the necessary Splunk processes and enables the server to accept incoming data. If the server is already running, this command will have no effect.

Splunkstart

```
root@coffely:/opt/splunk#./bin/splunk start
Splunk> Finding your faults, just like mom.
....
Checking prerequisites...
    Checking http port [8000]: open
    Checking mgmt port [8089]: open
    Checking appserver port [127.0.0.1:8065]: open
    Checking kvstore port [8191]: open
    Checking configuration... Done.
....
....
The Splunk web interface is at http://coffely:8000
```

As mentioned in the output, the Splunk dashboard will be accessible within the VM at `HTTP://coffely:8000`

Command: splunk stop

The `splunk stop` command is used to stop the Splunk server. This command stops all the running Splunk processes and disables the server from accepting incoming data. If the server is not running, this command will have no effect.

```
root@tryhackme:/opt/splunk#./bin/splunk stop
 ...some output ommitted ...
```

Command: splunk restart

The `splunk restart` command is used to restart the Splunk server. This command stops all the running Splunk processes and then starts them again. This is useful when changes have been made to the Splunk configuration files or when the server needs to be restarted for any other reason.

splunk: restart

```
root@tryhackme:/opt/splunk#./bin/splunk restart
...some output ommitted ...
```

Command: splunk status

The `splunk status` command is used to check the status of the Splunk server. This command will display information about the current state of the server, including whether it is running or not, and any errors that may be occurring.

```
root@coffely:/opt/splunk#./bin/splunk status
splunkd is running (PID: 2158).
splunk helpers are running (PIDs: 2159 2301 2351 2437).
```

Command: splunk add oneshot

The `splunk add oneshot` command is used to add a single event to the Splunk index. This is useful for testing purposes or for adding individual events that may not be part of a larger data stream.

```
root@coffely:/opt/splunk#./bin/splunk add oneshot
...some output ommitted ...
```

Command: splunk search

The `splunk search` command is used to search for data in the Splunk index. This command can be used to search for specific events, as well as to perform more complex searches using Splunk's search language.

```
root@coffely:/opt/splunk#./bin/splunk search coffely
WARNING: Server Certificate Hostname Validation is disabled. Please see
server.conf/[sslConfig]/cliVerifyServerName for details.
Feb 18 21:09:04 coffley ubuntu: coffely-has-the-best-coffee-in-town
Feb 18 13:48:17 coffely ubuntu: COFFELY
Feb 18 13:48:17 coffely ubuntu: COFFELY
```

Command: splunk help

The most important command is the help command which provides all the help options.

```
root@tryhackme:/opt/splunk#./bin/splunk help
Welcome to Splunk's Command Line Interface (CLI).

    Type these commands for more help:

    help [command]             type a command name to access its help page
    help [object]              type an object name to access its help page
    help [topic]               type a topic keyword to get help on a topic
    help commands              display a full list of CLI commands
    help clustering            commands that can be used to configure the
clustering setup
    help shclustering          commands that can be used to configure the
Search Head Cluster setup
    help control, controls     tools to start, stop, manage Splunk processes
    help datastore             manage Splunk's local filesystem use
    help distributed           manage distributed configurations such as
                               data cloning, routing, and distributed search
    help forwarding            manage deployments
    help input, inputs         manage data inputs
    help licensing             manage licenses for your Splunk server
    help settings              manage settings for your Splunk server
    help simple, cheatsheet    display a list of common commands with syntax
    help tools                 tools to help your Splunk server
    help search                help with Splunk searches
        ....
        ....
```

Install Forwarder

Change the user to sudo, unpack, and install the forwarder with the following command.

```
ubuntu@coffely:~/Downloads/splunk# sudo su
root@coffely:/home/ubuntu/Downloads/splunk# tar xvzf splunkforwarder.tgz
splunkforwarder/
splunkforwarder/swidtag/
splunkforwarder/swidtag/splunk-UniversalForwarder-primary.swidtag
splunkforwarder/ftr
splunkforwarder/openssl/
...
...
splunkforwarder/etc/deployment-apps/
splunkforwarder/etc/deployment-apps/README
splunkforwarder/etc/log-debug.cfg
```

The above command will install all required files in the folder `splunkforwarder`. Next, we will move this folder to `/opt/` path with the command `mv splunkforwarder /opt/`.

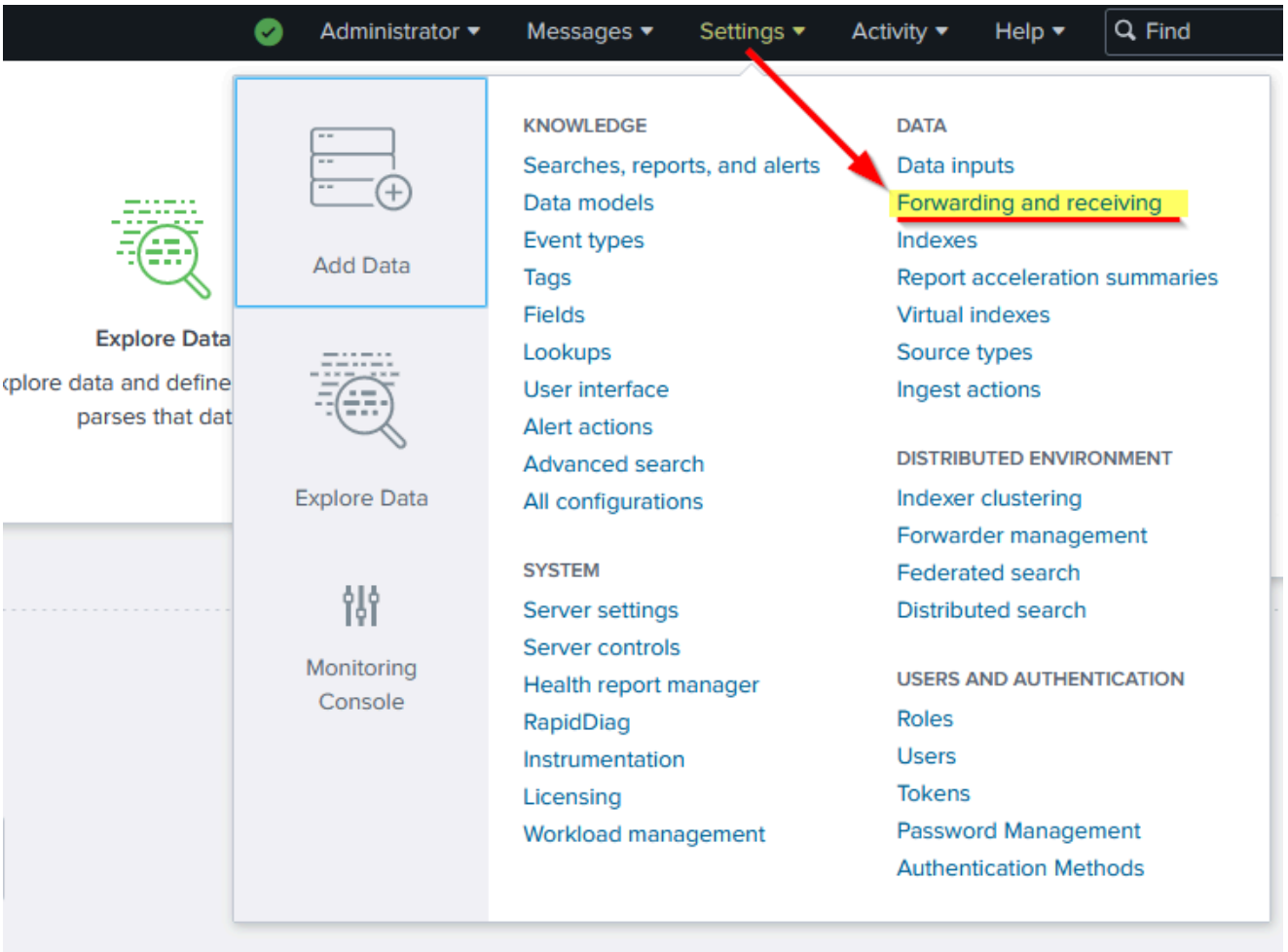
We will run the Splunk forwarder instance now and provide it with the new credentials as shown below:

Splunk Installation

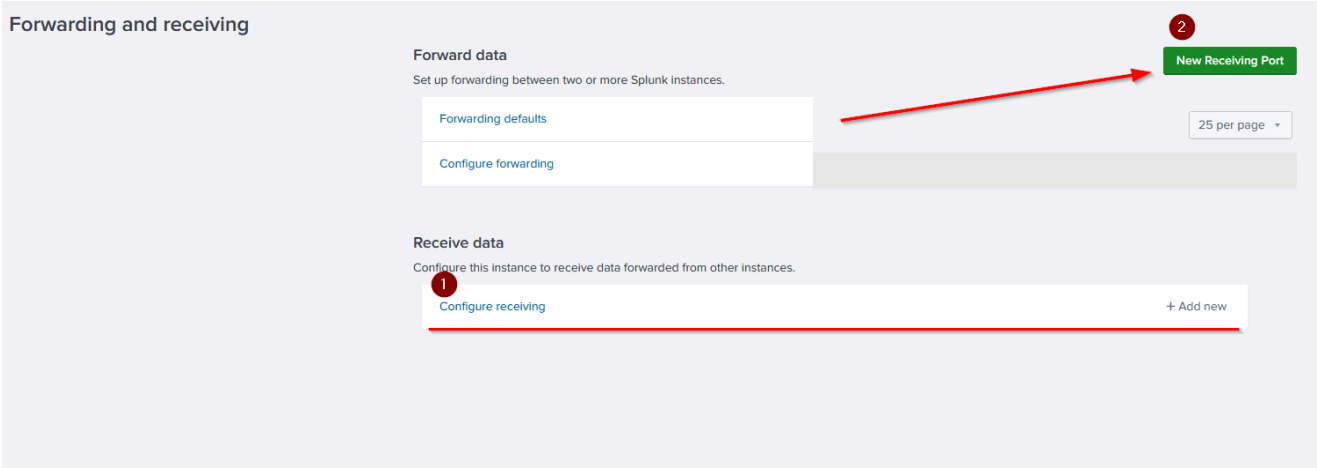
```
root@coffey:~/Downloads/splunk# mv splunkforwarder /opt/
root@coffey:~/Downloads/splunk# cd /opt/splunkforwarder
root@coffey:/opt/splunkforwarder# ./bin/splunk start --accept-license
This appears to be your first time running this version of Splunk.
...
...
Please enter an administrator username: splunkadmin
Password must contain at least:
   * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
Creating unit file...
Failed to auto-set default user.
...
...
Checking prerequisites...
    Checking mgmt port [8089]: not available
ERROR: mgmt port [8089] - port is already bound.  Splunk needs to use this
port.
Would you like to change ports? [y/n]: y
Enter a new mgmt port: 8090
Setting mgmt to port: 8090
The server's splunkd port has been changed.
    Checking mgmt port [8090]: open
Starting splunk server daemon (splunkd)...
Done
```

By default, **Splunk forwarder runs on port 8089.** If the system finds the port unavailable, it will ask the user for the custom port

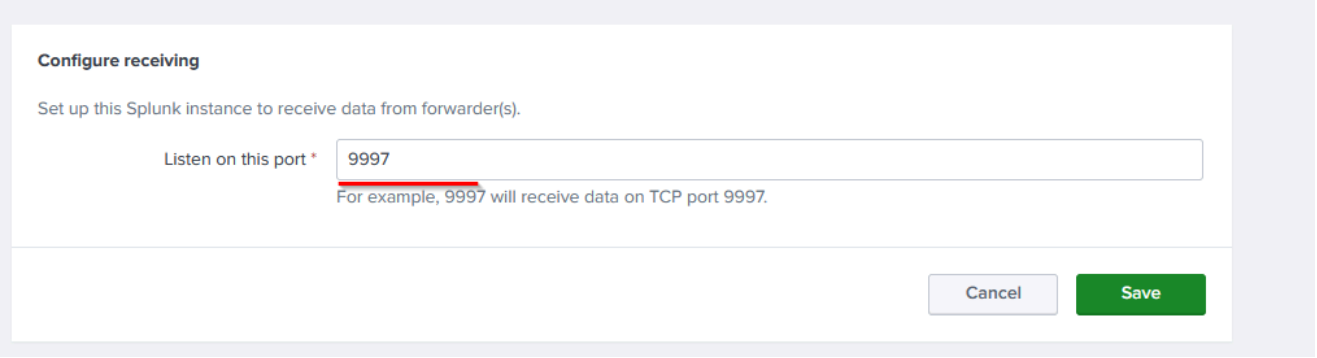
Splunk Configuration
Log into Splunk and Go to Settings -> Forward and receiving tab as shown below:



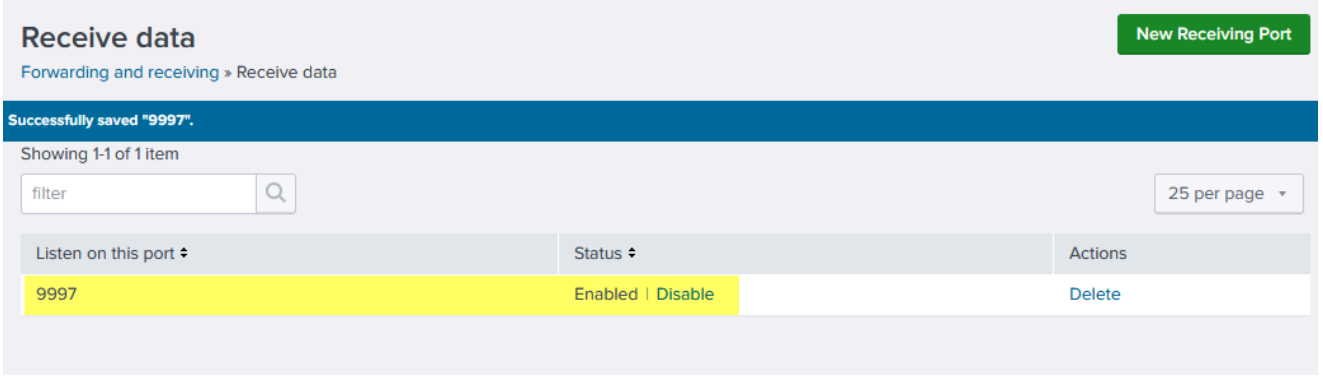
It will show multiple options to configure both forwarding and receiving. As we want to receive data from the Linux endpoint, we will click on **Configure receiving** and then proceed by configuring a new receiving port.



By default, the Splunk instance receives data from the forwarder on the port `9997`. It's up to us to use this port or change it. For now, we will configure our Splunk to start **listening on port 9997** and **Save**, as shown below:

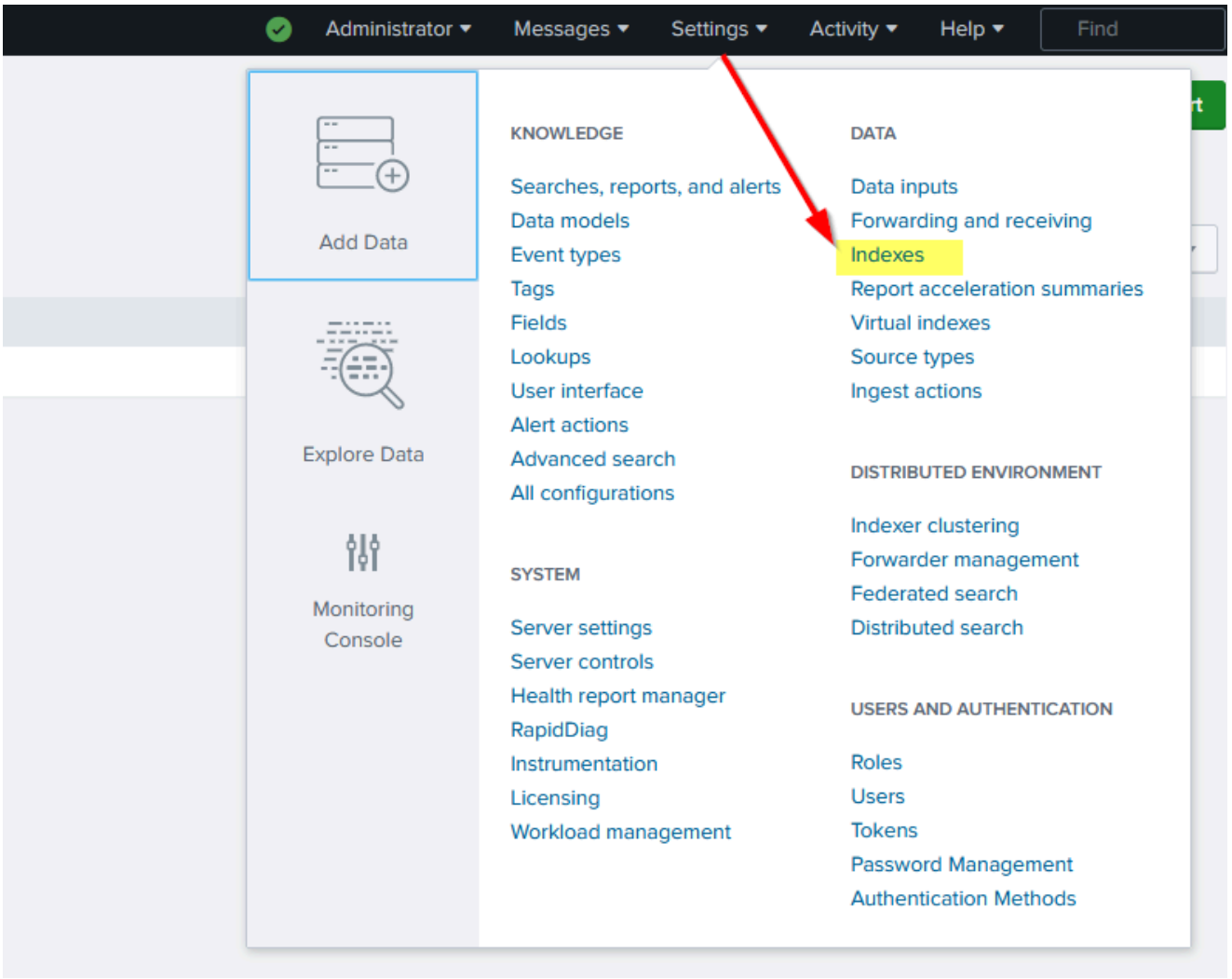


Our listening port 9997 is now enabled and waiting for the data. If we want, we can delete this entry by clicking on the `Delete` option under the `Actions` column.

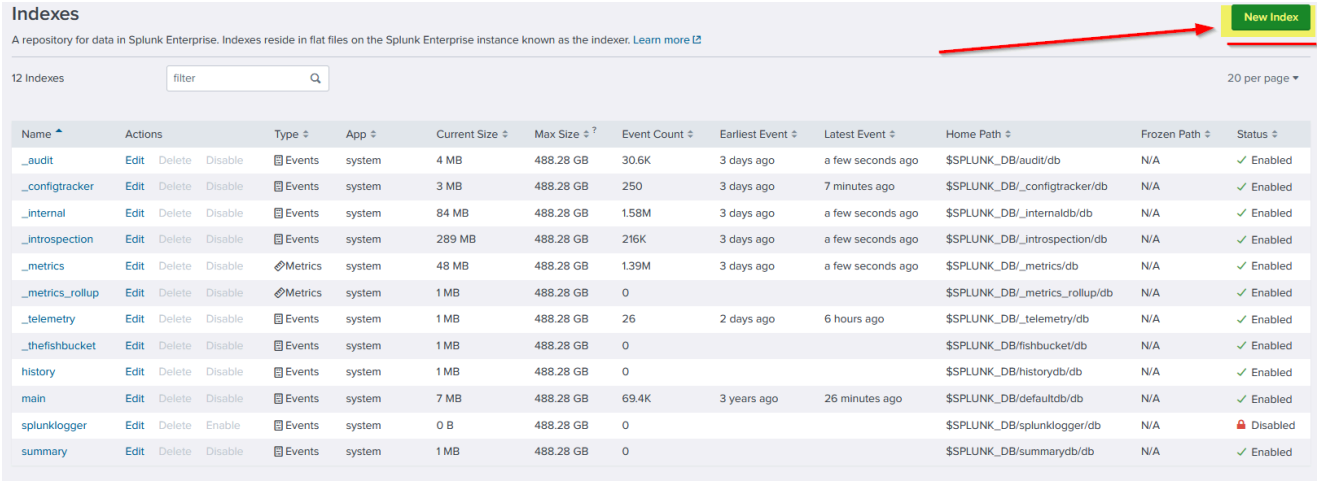


Creating Index

Now that we have enabled a listening port, the important next step is to create an index that will store all the receiving data. If we do not specify an index, it will start storing received data in the default index, which is called the `main` index.

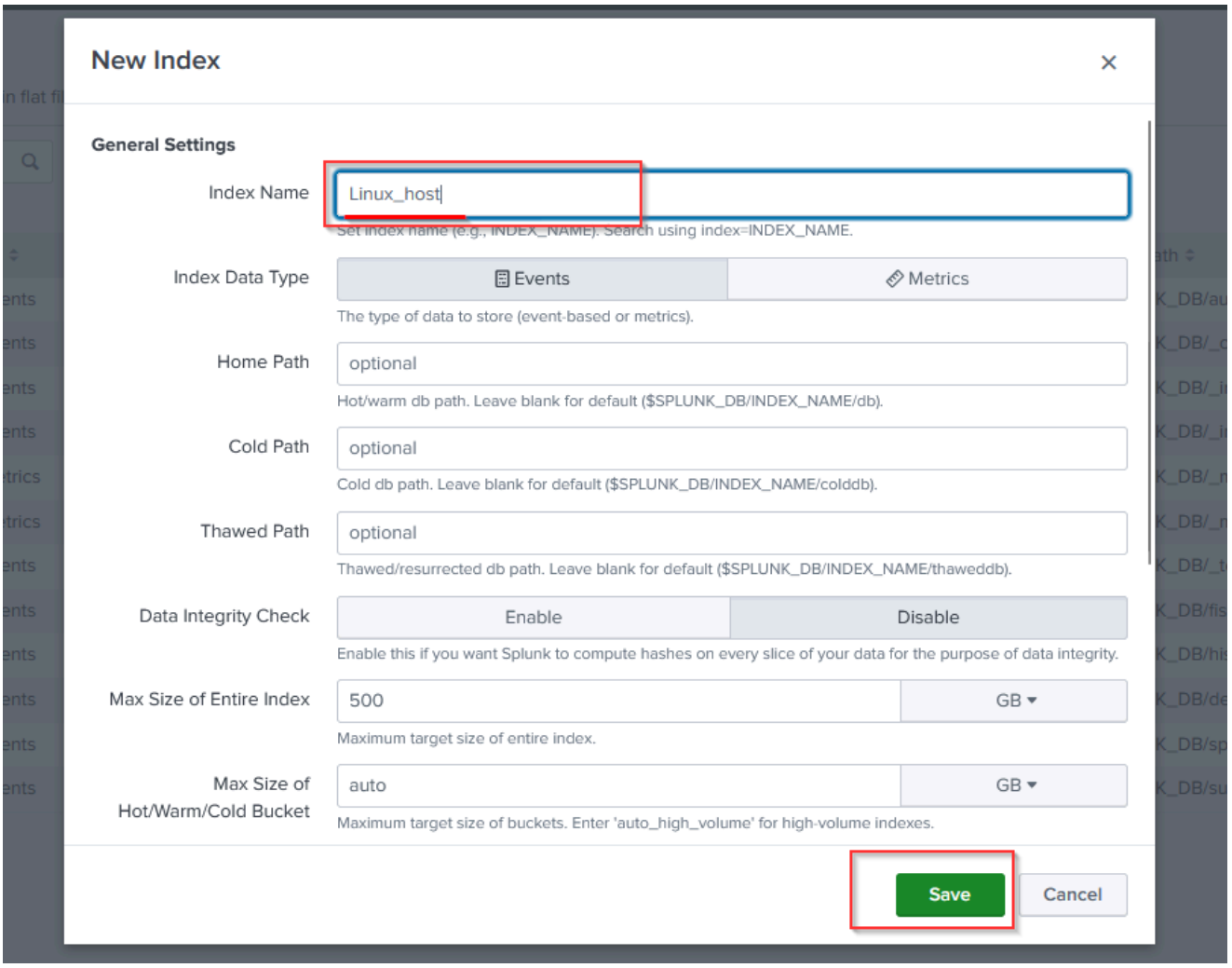


The indexes tab contains all the indexes created by the user or by default. This shows some important metadata about the indexes like Size, Event Count, Home Path, Status, etc.



Name	Actions	Type	App	Current Size	Max Size	Event Count	Earliest Event	Latest Event	Home Path	Frozen Path	Status
_audit	Edit Delete Disable	Events	system	4 MB	488.28 GB	30.6K	3 days ago	a few seconds ago	$SPLUNK_DB/audit/db	N/A	Enabled
_configtracker	Edit Delete Disable	Events	system	3 MB	488.28 GB	250	3 days ago	7 minutes ago	$SPLUNK_DB/_configtracker/db	N/A	Enabled
_internal	Edit Delete Disable	Events	system	84 MB	488.28 GB	1.58M	3 days ago	a few seconds ago	$SPLUNK_DB/_internaldb/db	N/A	Enabled
_introspection	Edit Delete Disable	Events	system	289 MB	488.28 GB	216K	3 days ago	a few seconds ago	$SPLUNK_DB/_introspection/db	N/A	Enabled
_metrics	Edit Delete Disable	Metrics	system	48 MB	488.28 GB	1.39M	3 days ago	a few seconds ago	$SPLUNK_DB/_metrics/db	N/A	Enabled
_metrics_rollup	Edit Delete Disable	Metrics	system	1 MB	488.28 GB	0			$SPLUNK_DB/_metrics_rollup/db	N/A	Enabled
_telemetry	Edit Delete Disable	Events	system	1 MB	488.28 GB	26	2 days ago	6 hours ago	$SPLUNK_DB/_telemetry/db	N/A	Enabled
_thefishbucket	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			$SPLUNK_DB/fishbucket/db	N/A	Enabled
history	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			$SPLUNK_DB/historydb/db	N/A	Enabled
main	Edit Delete Disable	Events	system	7 MB	488.28 GB	69.4K	3 years ago	26 minutes ago	$SPLUNK_DB/defaultdb/db	N/A	Enabled
splunklogger	Edit Delete Enable	Events	system	0 B	488.28 GB	0			$SPLUNK_DB/splunklogger/db	N/A	Disabled
summary	Edit Delete Disable	Events	system	1 MB	488.28 GB	0			$SPLUNK_DB/summarydb/db	N/A	Enabled

Click the **New Index** button, fill out the form, and click **Save** to create the index. Here we have created an index called `Linux_host` as shown below:



Configuring Forwarder

It's time to configure the forwarder to ensure it sends the data to the right destination. Back in the Linux host terminal, go to the `/opt/splunkforwarder/bin` directory:

Splunk: Forwarder

```
root@coffely:/opt/splunkforwarder/bin# ./splunk add forward-server
10.49.186.69:9997
WARNING: Server Certificate Hostname Validation is disabled. Please see
server.conf/[sslConfig]/cliVerifyServerName for details.
Splunk username: splunkadmin
Password:
Added forwarding to: 10.49.186.69:9997.
```

This command will add the forwarder server, which listens to port 9997.
Linux Log Sources
Linux stores all its important logs into the `/var/log` file, as shown below. In our case, we will ingest syslog into Splunk. All other logs can be ingested using the same method.

```
Xorg.0.log                  dmesg.1.gz                  prime-offload.log
Xorg.0.log.old              dmesg.2.gz                  prime-supported.log
alternatives.log            dmesg.3.gz                  private
amazon                      dmesg.4.gz                  samba
apport.log                  dpkg.log                    speech-dispatcher
apport.log.1                fontconfig.log              syslog
apt                         gdm3                        syslog.1
audit                       gpu-manager-switch.log      syslog.2.gz
auth.log                    gpu-manager.log             syslog.3.gz
auth.log.1                  hp                          syslog.4.gz
btmp                        journal                     syslog.5.gz
cloud-init-output.log       kern.log                    syslog.6.gz
cloud-init.log              kern.log.1                  syslog.7.gz
cups                        landscape                   unattended-upgrades
dist-upgrade                lastlog                     wtmp
dmesg                       lightdm
dmesg.0                     openvpn
```

Next, we will tell Splunk forwarder which logs files to monitor. Here, we tell Splunk Forwarder to monitor the `/var/log/syslog` file.

Ingest syslog file

```
root@coffely:/opt/splunkforwarder/bin# ./splunk add monitor /var/log/syslog
-index Linux_host
WARNING: Server Certificate Hostname Validation is disabled. Please see
server.conf/[sslConfig]/cliVerifyServerName for details.
Added monitor of '/var/log/syslog'.
```

Exploring Inputs.conf
We can also open the **inputs.conf** file located in `/opt/splunkforwarder/etc/apps/search/local`, and look at the configuration added after the commands we used above.

Inputs.conf

```
root@coffely:/opt/splunkforwarder/etc/apps/search/local# ls
inputs.conf
```

We can view the content of the `input.conf` using the cat command.

Inputs.conf

```
root@coffely:/opt/splunkforwarder/etc/apps/search/local# cat inputs.conf
[monitor:///var/log/syslog]
disabled = false
index = Linux_host
```

Utilizing Logger Utility

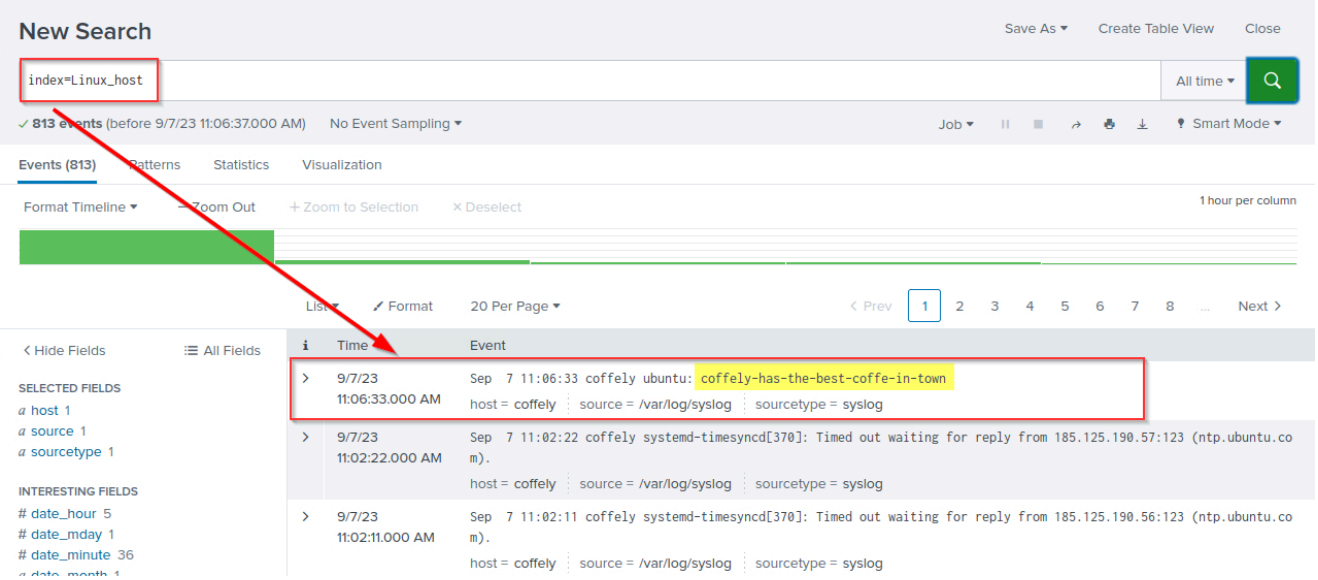
Logger is a built-in command line tool to create test logs added to the syslog file. As we are already monitoring the syslog file and sending all logs to the Splunk, the log we generate in the next step can be found with Splunk logs. To run the command, use the following command.

Logger: syslog

```
tryhackme@coffely:/opt/splunkforwarder/bin# logger "coffely-has-the-best-coffee-in-town"
```

Logger: syslog

```
tryhackme@coffely:/tryhackme@coffleylab:/opt/splunkforwarder/bin# tail -1 /var/log/syslog
```



Great, We have successfully installed and configured Splunk Forwarder to get the logs from the syslog file into Splunk.

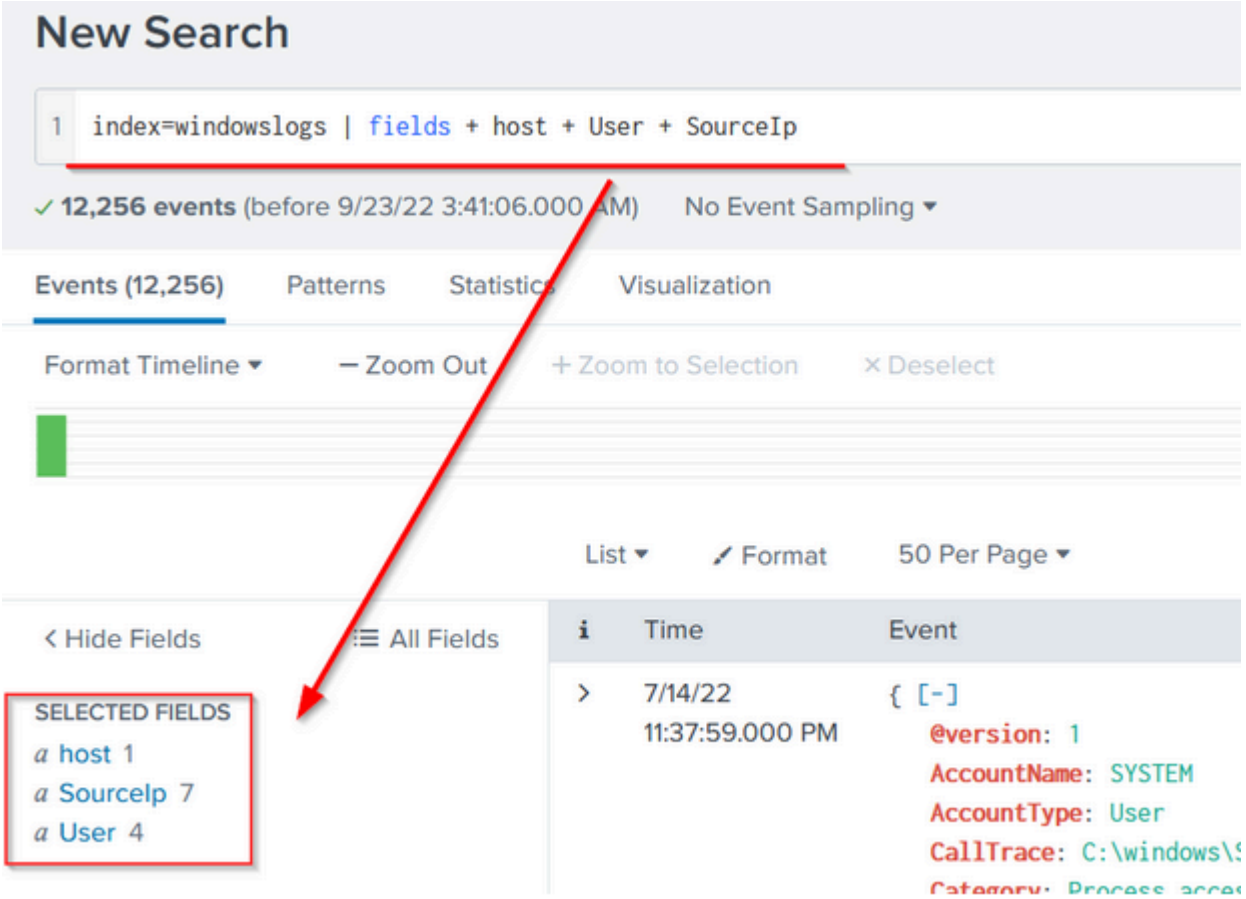
Filtering the Result in SPL(Search Processing Language)

Fields

Command	**fields**
Explanation	Fields command is used to add or remove mentioned fields from the search results. To remove the field, minus sign (-) is used before the fieldname and plus (+) is used before the fields which we want to display.
Syntax	| fields <field_name1> <field_name2>
Example	| `fields + HostName - EventID`

Let's use the fields command to only display host, User, and SourceIP fields using the following syntax.

Search Query: `index=windowslogs | fields + host + User + SourceIp`

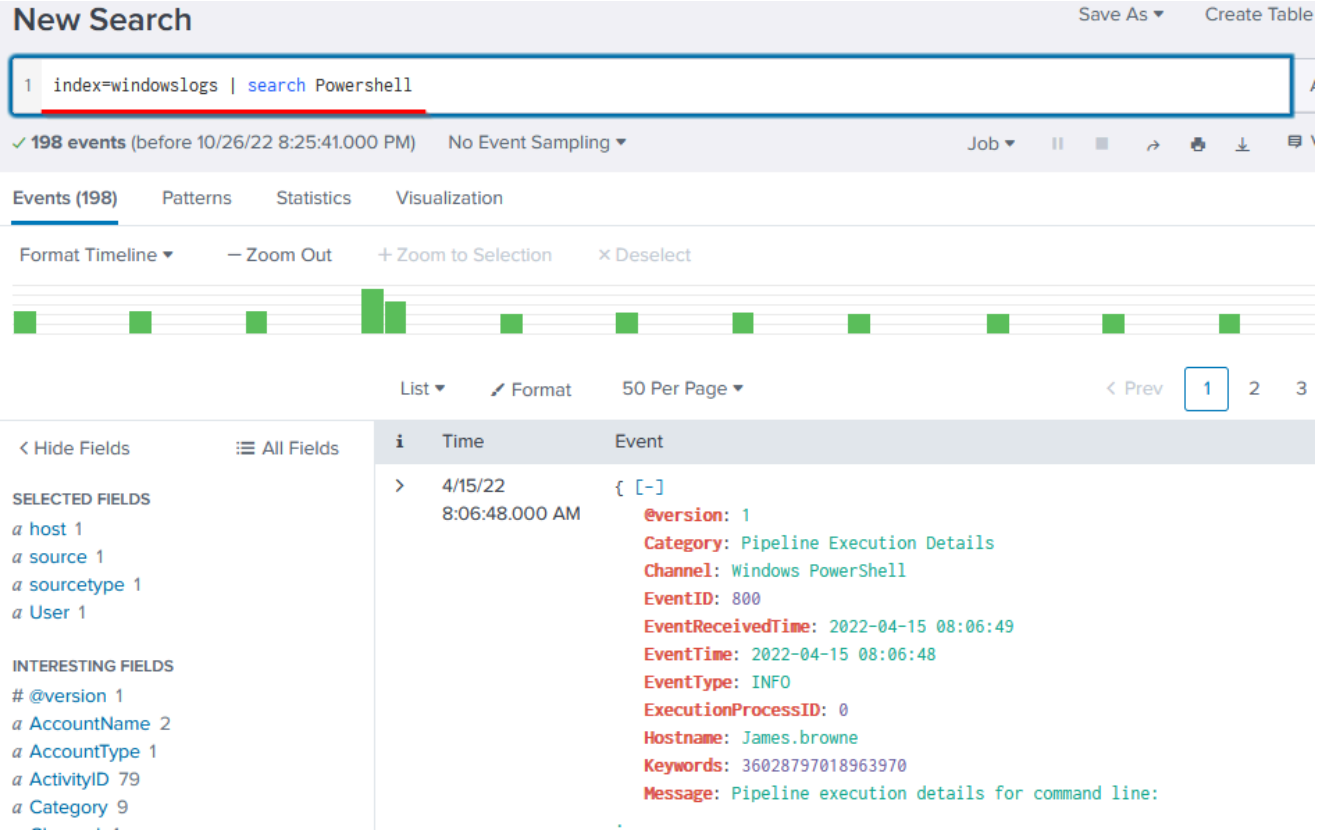


Search

Command	**search**
Explanation	This command is used to search for the raw text while using the chaining command `|`
Syntax	| search <search_keyword>
Example	`| search "Powershell"`

Use the search command to show all the events containing the term Powershell. This will return all the events that contain the term "**Powershell**".

Search Query: `index=windowslogs | search Powershell`

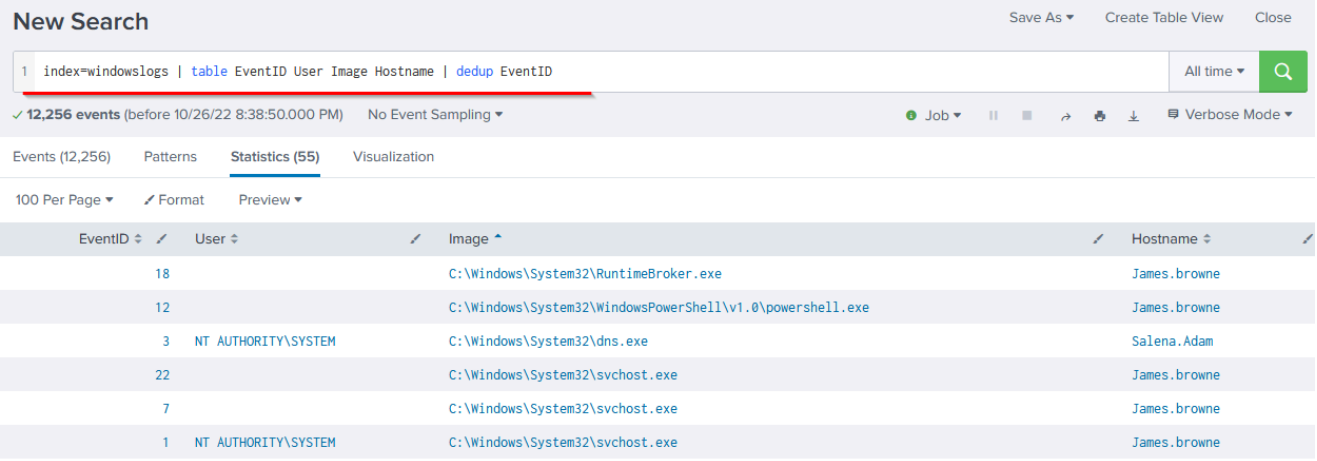


Dedup

Command	**dedup**
Explanation	Dedup is the command used to remove duplicate fields from the search results. We often get the results with various fields getting the same results. These commands remove the duplicates to show the unique values.
Syntax	| dedup <fieldname>
Example	`| dedup EventID`

We can use the dedup command to show the list of unique **EventIDs** from a particular hostname.

Search Query: `index=windowslogs | table EventID User Image Hostname | dedup EventID`



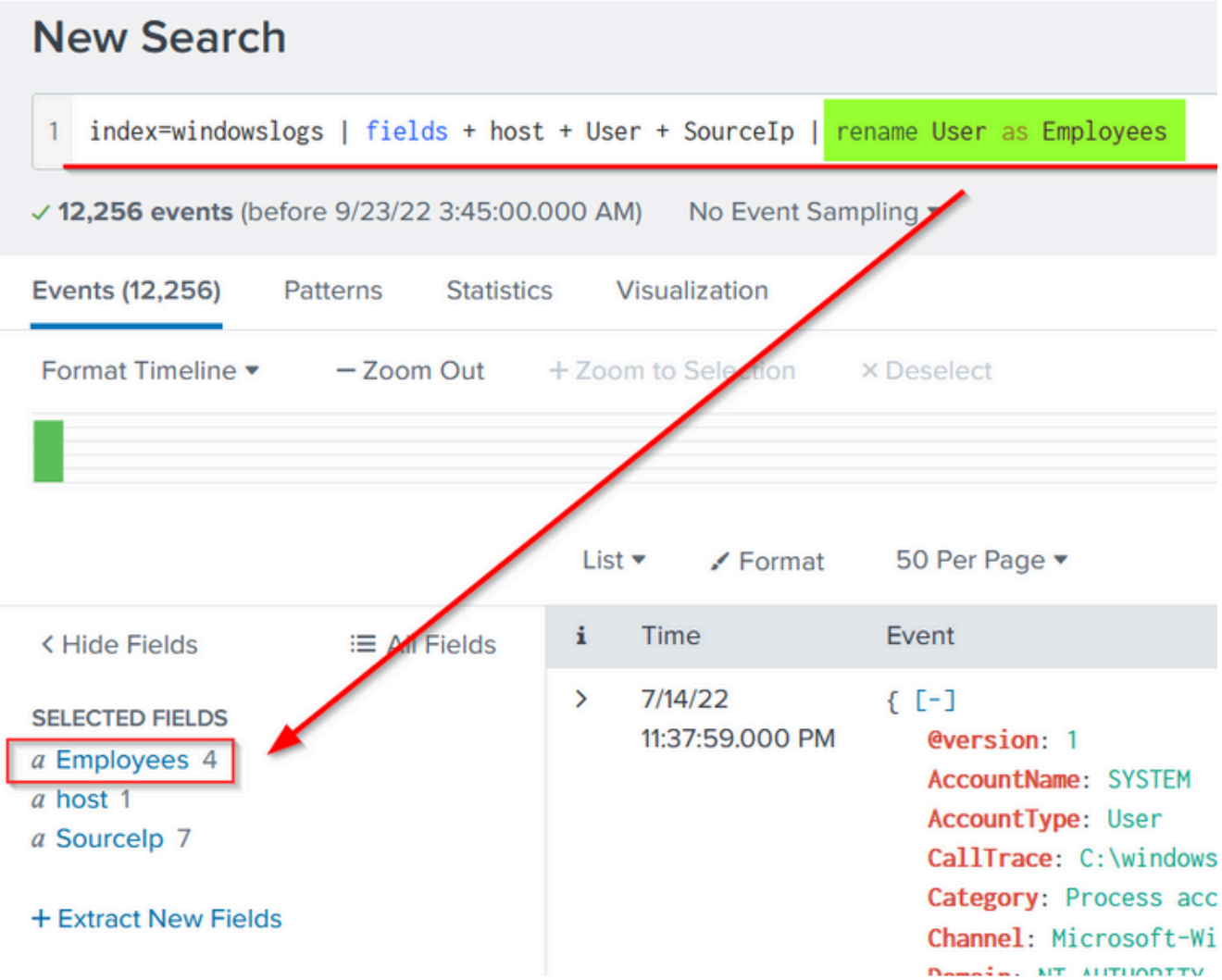
EventID	User	Image	Hostname
18		C:\Windows\System32\RuntimeBroker.exe	James.browne
12		C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	James.browne
3	NT AUTHORITY\SYSTEM	C:\Windows\System32\dns.exe	Salena.Adam
22		C:\Windows\System32\svchost.exe	James.browne
7		C:\Windows\System32\svchost.exe	James.browne
1	NT AUTHORITY\SYSTEM	C:\Windows\System32\svchost.exe	James.browne

Rename

Command	**rename**
Explanation	It allows us to change the name of the field in the search results. It is useful in a scenario when the field name is generic or log, or it needs to be updated in the output.
Syntax	| rename <fieldname>
Example	| `rename User as Employees`

Let's rename the User field to Employees using the following search query.

Search Query: `index=windowslogs | fields + host + User + SourceIp | rename User as Employees`



Structuring the Search Results

Table

Explanation	Each event has multiple fields, and not every field is important to display. The Table command allows us to create a table with selective fields as columns.
Syntax	| table <field_name1> <fieldname_2>
Example	| `table` | `head 20` # will return the top 20 events from the result list.

This search query will create a table with three columns selected and ignore all the remaining columns from the display.

Search Query: `index=windowslogs | table EventID Hostname SourceName`

New Search

```
1 index=windowslogs | table EventID Hostname SourceName
```

✓ **12,256 events** (before 10/26/22 9:07:01.000 PM) No Event Sampling

Events (12,256) Patterns **Statistics (12,256)** Visualization

100 Per Page Format Preview < Prev

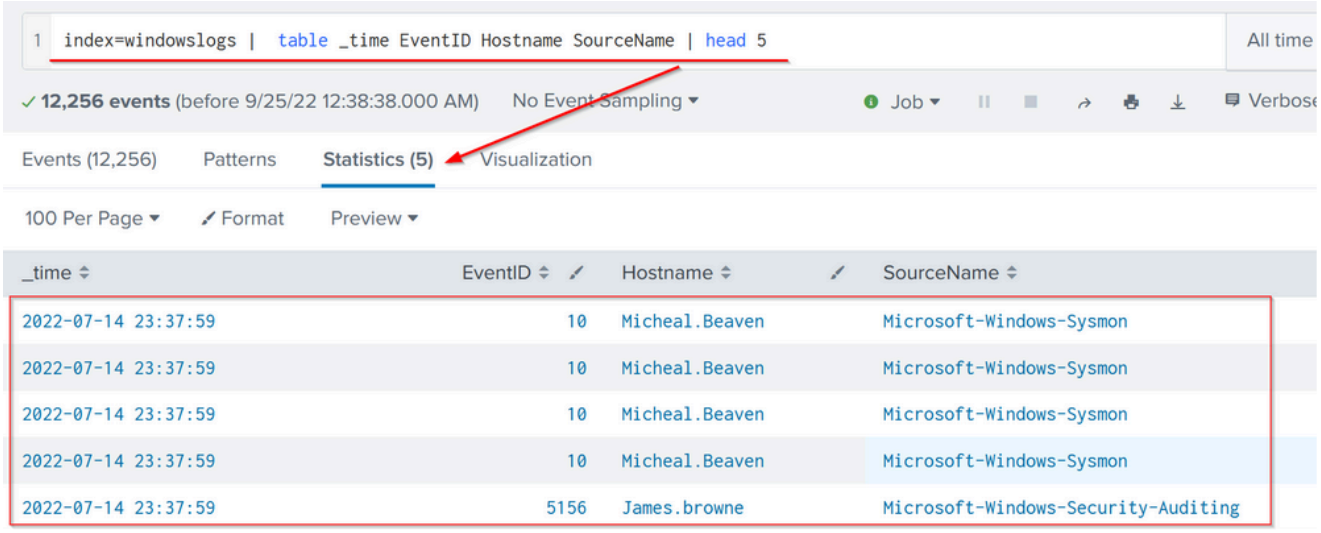
EventID	Hostname	SourceName
10	Micheal.Beaven	Microsoft-Windows-Sysmon
10	Micheal.Beaven	Microsoft-Windows-Sysmon
10	Micheal.Beaven	Microsoft-Windows-Sysmon
10	Micheal.Beaven	Microsoft-Windows-Sysmon
5156	James.browne	Microsoft-Windows-Security-Auditing
5158	James.browne	Microsoft-Windows-Security-Auditing
800	James.browne	PowerShell
4103	James.browne	Microsoft-Windows-PowerShell

Head

Explanation	The **head** command returns the first 10 events if no number is specified.
Syntax	| head <number>
Example	| `head` # will return the top 10 events from the result list | `head 20` # will return the top 20 events from the result list

The following search query will show the table containing the mentioned fields and display only the top 5 entries.

Search Query: `index=windowslogs | table _time EventID Hostname SourceName | head 5`



Tail

Explanation	The **Tail** command returns the last 10 events if no number is specified.
Syntax	| tail <number>
Example	| `tail` # will return the last 10 events from the result list | `tail 20` # will return the last 20 events from the result list

The following search query will show the table containing the mentioned fields and display only 5 entries from the bottom of the list.

Search Query: `index=windowslogs | table _time EventID Hostname SourceName | tail 5`

```
index=windowslogs | table _time EventID Hostname SourceName | tail 5
```

✓ **12,256 events** (before 9/25/22 1:14:18.000 AM) No Event Sampling ▾ Job ▾

Events (12,256) Patterns **Statistics (5)** Visualization

100 Per Page ▾ Format Preview ▾

_time	EventID	Hostname	SourceName
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800		PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell

sort

Explanation	The **Sort** command allows us to order the fields in ascending or descending order.
Syntax	| `sort` <field_name>
Example	| `sort Hostname` # This will sort the result in Ascending order.

The following search query will sort the results based on the Hostname field.

Search Query: `index=windowslogs | table _time EventID Hostname SourceName | sort Hostname`

```
1 index=windowslogs | table _time EventID Hostname SourceName | sort Hostname
```

✓ **12,256 events** (before 9/25/22 1:17:18.000 AM) No Event Sampling ▾ Job ▾

Events (12,256) Patterns **Statistics (10,000)** Visualization

100 Per Page ▾ Format Preview ▾ < Prev 1 2 3 4 5 6 7 8

_time	EventID	Hostname	SourceName
2022-07-14 23:37:59	5156	James.browne	Microsoft-Windows-Security-Auditing
2022-07-14 23:37:59	5158	James.browne	Microsoft-Windows-Security-Auditing
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell
2022-04-15 08:06:48	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:06:48	800	James.browne	PowerShell

Reverse

Explanation	The reverse command simply reverses the order of the events.
Syntax	| reverse
Example	`<Search Query> | reverse`

Search Query: `index=windowslogs | table _time EventID Hostname SourceName | reverse`

New Search

Save As ▾ Create

```
1 index=windowslogs | table _time EventID Hostname SourceName | reverse
```

✓ **12,256 events** (before 10/26/22 9:28:24.000 PM) No Event Sampling ▾ Job ▾

Events (12,256) Patterns **Statistics (12,256)** Visualization

100 Per Page ▾ Format Preview ▾ < Prev 1 2 3 4 5 6

_time	EventID	Hostname	SourceName
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800		PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103		Microsoft-Windows-PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	800	James.browne	PowerShell
2022-04-15 08:05:46	4103	James.browne	Microsoft-Windows-PowerShell

Transformational Commands in SPL

Top

Command	**top**
Explanation	This command returns frequent values for the top 10 events.
Syntax	| top <field_name> | top limit=6 <field_name>
Example	`top limit=3 EventID`

The following command will display the top 7 Image (representing Processes) captured.

Search Query: `index=windowslogs | top limit=7 Image`

```
index=windowslogs | top limit=7 Image
```

All time

✓ **12,256 events** (before 6/28/22 11:29:39.000 PM) No Event Sampling ▾ Job ▾ Verbose Mode ▾

Events (12,256) Patterns **Statistics (7)** Visualization

100 Per Page ▾ Format Preview ▾

Image	count	percent
C:\windows\system32\svchost.exe	1642	38.836329
C:\windows\system32\backgroundTaskHost.exe	547	12.937559
C:\Windows\System32\svchost.exe	426	10.075686
C:\windows\system32\taskhostw.exe	250	5.912961
C:\Windows\System32\BackgroundTransferHost.exe	210	4.966887
C:\Windows\System32\backgroundTaskHost.exe	196	4.635762
C:\Windows\System32\wbem\WmiPrvSE.exe	108	2.554399

Rare

Command	**rare**
Explanation	This command does the opposite of top command as it returns the least frequent values or bottom 10 results.
Syntax	| rare <field_name> | rare limit=6 <field_name>
Example	`rare limit=3 EventID`

The following command will display the rare 7 Image (Processes) captured.

Search Query: `index=windowslogs | rare limit=7 Image`

```
index=windowslogs | rare limit=7 Image
```

All time

✓ **12,256 events** (before 6/28/22 11:32:48.000 PM) No Event Sampling ▾ Job ▾ Verbose Mode ▾

Events (12,256) Patterns **Statistics (7)** Visualization

100 Per Page ▾ Format Preview ▾

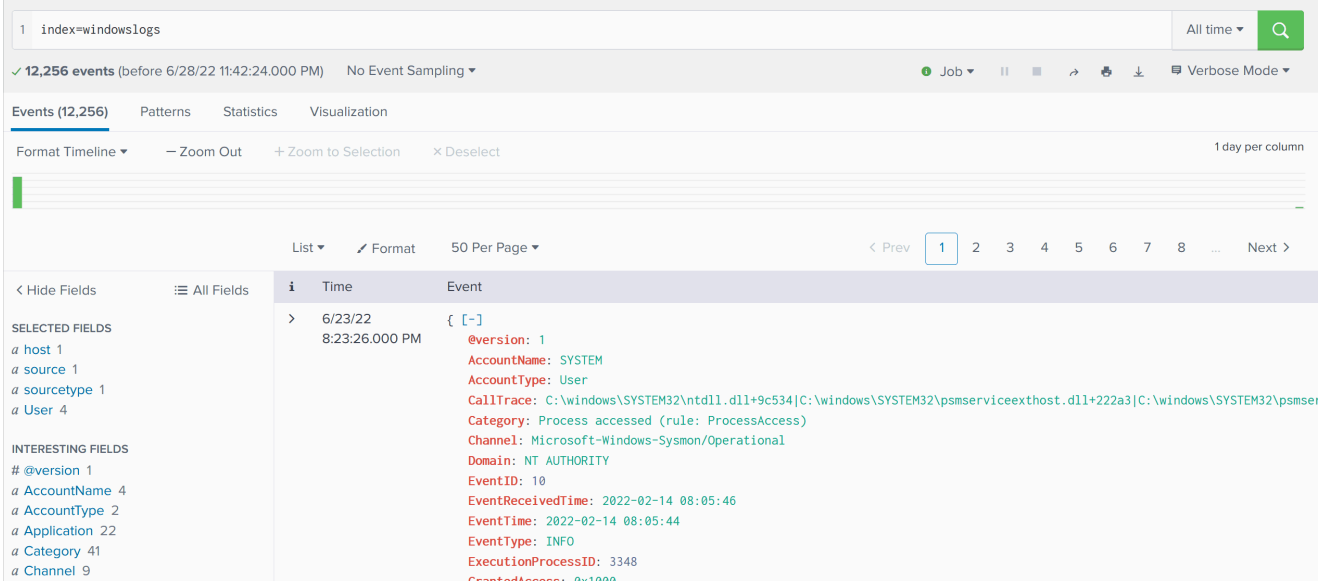
Image	count	percent
C:\Windows\ADWS\Microsoft.ActiveDirectory.WebServices.exe	1	0.023652
C:\windows\system32\SecurityHealthService.exe	1	0.023652
C:\windows\system32\net1.exe	1	0.023652
C:\windows\ADWS\Microsoft.ActiveDirectory.WebServices.exe	2	0.047304
C:\Windows\System32\dfsrs.exe	3	0.070956
C:\windows\Explorer.EXE	3	0.070956
C:\windows\system32\services.exe	5	0.118259

Highlight

Command	**highlight**
Explanation	The highlight command shows the results in raw events mode with fields highlighted.
Syntax	highlight <field_name1> <field_name2>
Example	`highlight User, host, EventID, Image`

The following command will highlight the three mentioned fields in the raw logs

Search Query: `index=windowslogs | highlight User, host, EventID, Image`



STATS Commands

SPL supports various stats commands that help in calculating statistics on the values. Some common stat commands are:

Command	Explanation	Syntax	Example
Average	This command is used to calculate the average of the given field.	stats avg(field_name)	stats avg(product_price)
Max	It will return the maximum value from the specific field.	stats max(field_name)	stats max(user_age)
Min	It will return the minimum value from the specific field.	stats min(field_name)	stats min(product_price)
Sum	It will return the sum of the fields in a specific value.	stats sum(field_name)	stats sum(product_cost)
Count	The count command returns the number of data occurrences.	stats count(function) AS new_NAME	stats count(source_IP)

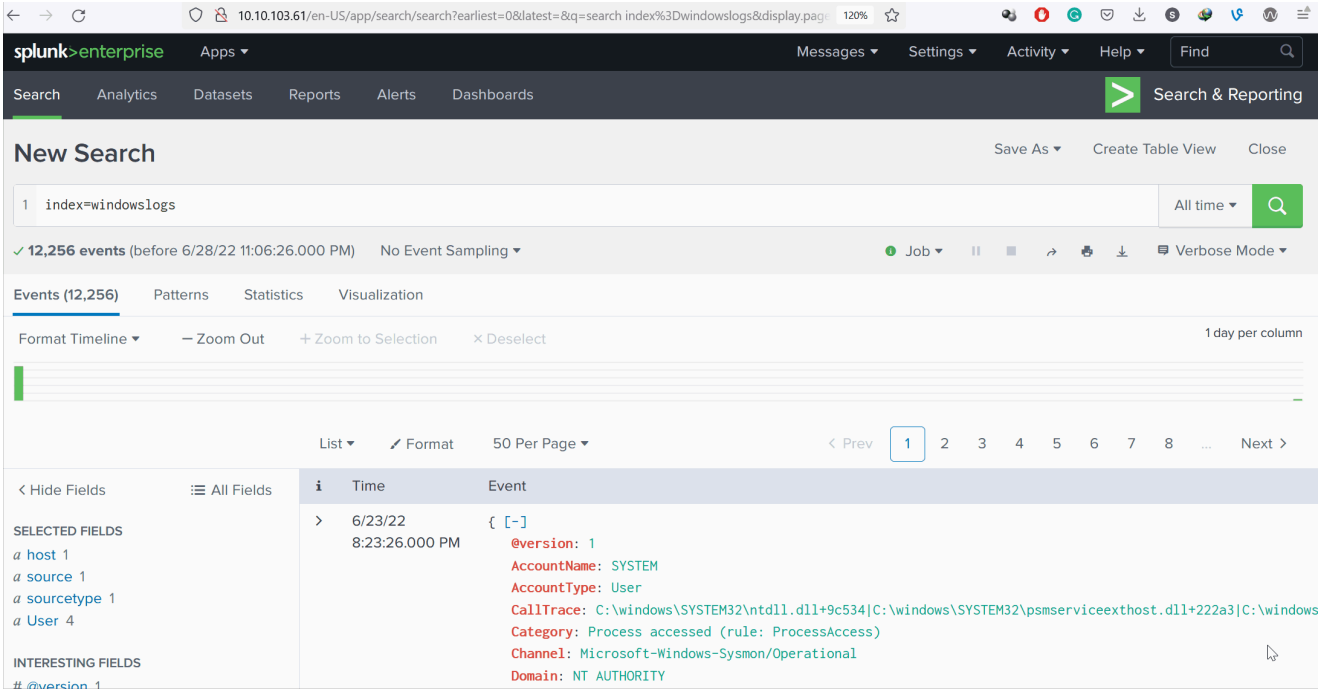
Splunk Chart Commands

These are very important types of transforming commands that are used to present the data in table or visualization form. Most of the chart commands utilize various stat commands.

Chart

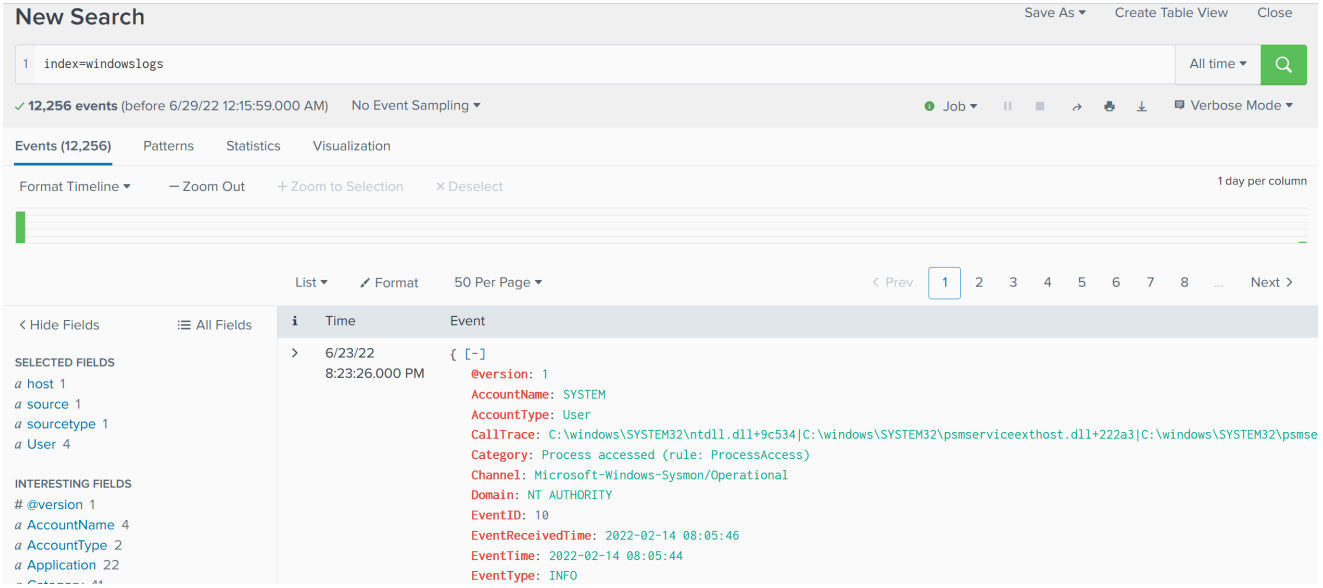
Command	**chart**
Explanation	The chart command is used to transform the data into tables or visualizations.
Syntax	| chart <function>
Example	`| chart count by User`

Search Query: `index=windowslogs | chart count by User`



Timechart

Command	**timechart**
Explanation	The timechart command returns the time series chart covering the field following the function mentioned. Often combined with STATS commands.
Syntax	| timechart function <field_name>
Example	`| timechart count by Image`

The following query will display the Image chart based on the time.

Search Query: `index=windowslogs | timechart count by Image`